A certified zero-free region for the Riemann zeta function in the half-plane $\Re s \geq 0.6$

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Abstract

We establish a certified zero-free region for the Riemann zeta function: $\zeta(s) \neq 0$ for $\Re s \geq 0.6$. On $\Omega = \{\Re s > \frac{1}{2}\}$ we introduce an arithmetic ratio $\mathcal{J}(s)$ and the associated Cayley transform $\Theta(s) = (2\mathcal{J}(s) - 1)/(2\mathcal{J}(s) + 1)$. A boundary wedge condition (P^+) implies that Θ is Schur on Ω, which in turn yields a half-plane Schur/Herglotz removability argument excluding poles of $\mathcal{J}$ throughout $\{\Re s \geq 0.6\}$. Since any pole of $\mathcal{J}$ in Ω forces a zero of ζ, the pole exclusion gives the stated zero-free region. Supplementary rigorous ball-arithmetic certificates are included in the handoff bundle for independent reproducibility checks on representative low-height rectangles.

1 Introduction

The Riemann zeta function

$$\zeta(s) \;=\; \sum_{n\geq 1} \frac{1}{n^s}, \qquad \Re s > 1,$$

extends meromorphically to $\mathbb{C}$ with a simple pole at $s = 1$ and satisfies a functional equation after completion. Its nontrivial zeros govern the finest fluctuations in the distribution of prime numbers, and the Riemann Hypothesis (RH) asserts that all such zeros lie on the critical line $\Re s = \frac{1}{2}$; see [1, 3] for background.

This paper isolates an unconditional, fixed half-plane statement in the direction of RH. Unlike classical zero-free regions near $\Re s = 1$ (which are asymptotic in height), the result here is a *uniform* half-plane exclusion at $\Re s \geq 0.6$.

Theorem 1 (Certified far-field zero-freeness)**.** *The Riemann zeta function has no zeros in the region* $\{\, s \in \mathbb{C} : \; \Re s \geq 0.6 \,\}$.

Scope and supplementary computation. The proof of Theorem 1 is analytic and self-contained in the present text, with the boundary wedge certificate (P+) proved in Appendix A. The shipped computational artifacts provide independently checkable ball-arithmetic cross-checks on representative low-height rectangles (Table 1) together with a verification protocol (Appendix B). They are included for transparency and reproducibility, but are not logically required for the all-heights implication chain in the proof.

The Appendix records the analytic function theory inputs used in the proof of (P+) (boundary traces, factorization, and the phase–velocity identity).

Strategy: Schur pinching via a Cayley field

We work on the right half-plane $\Omega = \{\, \Re s > \frac{1}{2} \,\}$. In Section 2 we define an arithmetic ratio $\mathcal{J}$ (in the default *raw ζ-gauge*) with the following two structural properties:

- **(normalization at $+\infty$)** $\mathcal{J}(\sigma + it) \to 1$ as $\sigma \to +\infty$, hence $\Theta(\sigma + it) \to \frac{1}{3}$ (Remark 3);
- **(normalization)** $\det_2(I - A(s))$ is holomorphic and nonvanishing on Ω, so any zero of ζ in Ω produces a pole of $\mathcal{J}$ (Lemma 4).

We then pass to the Cayley transform

$$\Theta(s) \;:=\; \frac{2\mathcal{J}(s) - 1}{2\mathcal{J}(s) + 1}.$$

The analytic mechanism is a *Schur/Herglotz pinch* proved in Section 3: if Θ is Schur on a domain (i.e. $|\Theta| \le 1$) and not identically 1, then boundedness forces removability of any isolated singularity and prevents poles of $\mathcal{J}$. Since $\Theta(\sigma + it) \to \frac{1}{3}$ as $\sigma \to +\infty$, the degenerate possibility $\Theta \equiv 1$ is excluded on the half-planes relevant here. Therefore, to prove Theorem 1 it suffices to certify a Schur bound for the default Cayley field Θ_{raw} on some open half-plane $\{\, \Re s > 0.6 - \varepsilon \,\}$.

Certified inputs (what is rigorously checked)

The logical implication of Theorem 1 rests on a boundary certificate: we establish a boundary wedge (P+) (Lebesgue-a.e.) for the boundary phase of $\mathcal{J}$ on $\Re s = \frac{1}{2}$, which implies that $2\mathcal{J}$ is Herglotz and Θ is Schur on $\Omega \setminus Z(\zeta)$. The Schur/Herglotz pinch mechanism then excludes poles of $\mathcal{J}$ on $\{\, \Re s \ge 0.6 \,\}$ and hence excludes zeros of ζ there.

Supplementary computational cross-checks (not used in the proof). The handoff bundle also contains rigorous ball-arithmetic rectangle checks and finite Pick artifacts on low-height regions; these are included as independent numerical corroboration but are not used in the all-heights proof.

Reproducibility and verification materials

The handoff bundle (and repository) includes: (i) the verifier script based on ARB ball arithmetic (`python-flint`), and (ii) the JSON artifacts that record the certified maxima, spectral separation bounds, and denominator checks for the supplementary certificates listed in Table 1. The included `MANIFEST.md` and Appendix B provide a verification manifest mapping certificates to exact commands and expected output fields.

Place in a series

This paper is designed to stand alone as an unconditional certified zero-free region. Two companion papers (not required for Theorem 1) treat: (a) effective near-field energy barriers and Carleson budgets, and (b) a cutoff principle yielding conditional closure of RH.

The remainder of the paper defines the arithmetic ratio $\mathcal{J}$ and Cayley field Θ, proves the Schur pinch mechanism, and then discharges the Schur bound via the hybrid certification outlined above.

2 Definitions and main objects

This section defines the analytic objects used throughout the proof and records the basic relationships between zeros of ζ and the bounded-real (Schur/Herglotz) structure employed later. All definitions here are standard; precise trace-ideal references are cited when regularized determinants appear.

The completed zeta function and the far half-plane

Let $\zeta(s)$ denote the Riemann zeta function. We write $\xi(s)$ for the completed zeta function

$$\xi(s) \;:=\; \tfrac{1}{2}\, s(s-1)\, \pi^{-s/2}\Gamma(s/2)\, \zeta(s),$$

which is entire and satisfies the functional equation $\xi(s) = \xi(1-s)$; see [3]. In this paper, when we say "zero" we mean a zero of ζ (equivalently of ξ away from the canceled singularities at $s = 0, 1$) lying in the half-plane

$$\Omega \;:=\; \{\, s \in \mathbb{C} :\; \Re s > \tfrac{1}{2} \,\}.$$

Theorem 1 concerns the fixed far region $\{\, \Re s \geq 0.6 \,\} \subset \Omega$.

The prime-diagonal operator and the regularized determinant

Let $\mathcal{P}$ denote the set of primes and write $\ell^2(\mathcal{P})$ for the Hilbert space with orthonormal basis $\{e_p\}_{p\in\mathcal{P}}$. For $s \in \mathbb{C}$ define the prime-diagonal operator

$$A(s) : \ell^2(\mathcal{P}) \to \ell^2(\mathcal{P}), \qquad A(s)e_p := p^{-s} e_p.$$

For $\Re s > 1/2$,

$$\|A(s)\|_{\mathrm{HS}}^2 = \sum_{p\in\mathcal{P}} |p^{-s}|^2 = \sum_{p\in\mathcal{P}} p^{-2\Re s} \leq \sum_{n\geq 2} n^{-2\Re s} < \infty,$$

so $A(s)$ is Hilbert–Schmidt on Ω. In particular, the regularized determinant $\det_2(I - A(s))$ is well-defined and holomorphic on Ω (see [5, Ch. III] and [7, Ch. 9]).

Lemma 2 (Diagonal product formula for $\det_2$)**.** *Let T be a diagonal Hilbert–Schmidt operator on ℓ^2 with eigenvalues $\{\lambda_n\}$ satisfying $\sum_n |\lambda_n|^2 < \infty$. Then*

$$\det_2(I - T) \;=\; \prod_n (1 - \lambda_n)\, e^{\lambda_n},$$

where the product converges absolutely. In particular, $\det_2(I - T) = 0$ *iff* $\lambda_n = 1$ *for some* n.

Proof. This is standard for the $\mathcal{S}_2$-regularized determinant; see [5, Ch. III] or [7, Ch. 9]. (We only use the diagonal case and the zero criterion $\lambda_n = 1$.) □

Applying Lemma 2 to $T = A(s)$ on Ω gives the explicit product

$$\det_2(I - A(s)) \;=\; \prod_{p\in\mathcal{P}} (1 - p^{-s})\, e^{p^{-s}}. \tag{1}$$

The region $\Omega \subset \{\Re s > 1/2\}$ lies away from $s = 0$, so the compensator $1/s$ introduces no pole on the working domain. The point $s = 1$ lies in Ω, but the factor $(s-1)$ cancels the simple pole of ζ there. All holomorphy/pole assertions for $\mathcal{J}$ are made only on Ω, and poles are tracked relative to zeros of ζ in Ω.

Since $\Re s > 1/2$ implies $|p^{-s}| < 1$ for every prime p, each factor in (1) is nonzero. Hence $\det_2(I - A(s))$ is holomorphic and zero-free on Ω.

The arithmetic ratio $\mathcal{J}$ and the Cayley field Θ

Fix a domain $D \subset \Omega$. To allow numerically stable certified bounds later, we permit a holomorphic nonvanishing *normalizer* (or *gauge*) $\mathcal{O}$ on D, and define

$$\mathcal{J}(s) \;:=\; \frac{\det_2(I - A(s))}{\zeta(s)} \cdot \frac{s-1}{s} \cdot \frac{1}{\mathcal{O}(s)}, \qquad s \in D. \tag{2}$$

The factor $(s-1)$ cancels the simple pole of ζ at $s = 1$; the factor $1/s$ plays no role on $D \subset \Omega$ (but is convenient in later normalization). Unless explicitly stated otherwise, we work in the *raw ζ-gauge* $\mathcal{O} \equiv 1$ and denote the resulting objects by $\mathcal{J}_{\mathrm{raw}}$ and Θ_{raw}; for readability we usually drop the subscript in this default gauge.

Remark 3 (Gauge changes and what they do *not* change). If $\mathcal{O}$ is holomorphic and nonvanishing on D, then multiplying by $\mathcal{O}^{-1}$ cannot introduce poles on D. Thus the pole set of $\mathcal{J}$ on D is independent of the choice of gauge. However, quantitative bounds (e.g. Schur bounds on Θ) are not gauge-invariant; when a nontrivial gauge is used for a certified bound, one must also certify that $\mathcal{O}$ is holomorphic and nonvanishing on the certified domain. In the raw gauge $\mathcal{O} \equiv 1$ one has $\mathcal{J}(s) \to 1$ as $\Re s \to +\infty$, and hence $\Theta(s) \to 1/3$.

The associated Cayley transform is

$$\Theta(s) \;:=\; \frac{2\mathcal{J}(s) - 1}{2\mathcal{J}(s) + 1}. \tag{3}$$

Heuristically, $2\mathcal{J}$ plays the role of a Herglotz-type quantity and Θ the corresponding Schur function. The next section makes precise the key implication used later: a Schur bound on Θ forces removability of any isolated singularity of $\mathcal{J}$.

Lemma 4 (Zeros of ζ produce poles of $\mathcal{J}$)**.** *Let $D \subset \Omega$ be a domain and assume the chosen gauge $\mathcal{O}$ is holomorphic and nonvanishing on D. If $\rho \in D$ is a zero of $\zeta(s)$, then ρ is a pole of $\mathcal{J}(s)$ defined in* (2)*.*

Proof. By (2), the only possible singularities of $\mathcal{J}$ on D arise from zeros of ζ and from zeros of $\mathcal{O}$. The latter do not occur by assumption. The factor $(s-1)/s$ is holomorphic and nonzero on $D \subset \Omega$. Finally, $\det_2(I - A(s))$ is holomorphic and nonzero on Ω by (1). Hence a zero of ζ at ρ forces a pole of $\mathcal{J}$ at ρ. □

Schur and Herglotz classes (terminology)

Let $D \subset \mathbb{C}$ be a domain. A holomorphic function Θ on D is called *Schur* if $|\Theta| \le 1$ on D. A holomorphic function H on D is called *Herglotz* if $\Re H \ge 0$ on D. The Cayley transform identifies these classes: if H is Herglotz and $H \not\equiv -1$, then

$$\Theta = \frac{H-1}{H+1}$$

is Schur. Conversely, if Θ is Schur and $\Theta \not\equiv 1$, then $(1+\Theta)/(1-\Theta)$ is Herglotz; see [5, 6].

Outline of the far-field strategy in this language

Theorem 1 will follow once we establish that Θ is Schur on $\{\, \Re s > 0.6 \,\}$ (in the relevant gauge). Indeed, if $|\Theta| \le 1$ holds on $\{\, \Re s > 0.6 \,\}$ away from the poles of $\mathcal{J}$, then Θ is bounded near any isolated singularity, and thus extends holomorphically across it by the removable singularity theorem. Since poles of $\mathcal{J}$ correspond to zeros of ζ in Ω by Lemma 4, this rules out zeros of ζ in the far region. The precise pinch argument is proved in the next section.

3 Schur/Herglotz pinch mechanism

This section records the analytic mechanism that converts a Schur bound for the Cayley field Θ into a zero-free region for ζ. The point is that a holomorphic function bounded by 1 cannot have a pole, and any isolated singularity is removable. In our setting, poles of $\mathcal{J}$ in Ω encode zeros of ζ (Lemma 4), so a Schur bound forces those zeros to be absent.

Removable singularities under a Schur bound

Lemma 5 (Removable singularity under a strict Schur bound)**.** *Let $D \subset \mathbb{C}$ be a disc centered at ρ and let Θ be holomorphic on $D \setminus \{\rho\}$ with $|\Theta(s)| < 1$ there. Then Θ extends holomorphically to D and satisfies $|\Theta(s)| < 1$ for all $s \in D$. Consequently, the Cayley inverse*

$$\mathcal{H}(s) \;:=\; \frac{1+\Theta(s)}{1-\Theta(s)}$$

is holomorphic on D and satisfies $\Re\mathcal{H}(s) > 0$ on D.

Proof. Since Θ is bounded on the punctured disc, Riemann's removable singularity theorem gives a holomorphic extension to D (e.g. [8, Ch. 2]). The extension still satisfies $|\Theta| \le 1$ on D by continuity. If $|\Theta(\rho)| = 1$, then $|\Theta|$ attains its maximum at an interior point, so Θ is constant unimodular on D by the Maximum Modulus Principle (e.g. [8, Ch. 2]); this contradicts the strict bound $|\Theta| < 1$ on $D \setminus \{\rho\}$. Hence $|\Theta(\rho)| < 1$, and therefore $|\Theta| < 1$ holds everywhere on D. The Möbius map $w \mapsto (1+w)/(1-w)$ sends the unit disc into the right half-plane, so $\Re\mathcal{H} > 0$ on D. □

From a Schur bound to absence of poles

Corollary 6 (Schur bound prevents poles of $\mathcal{J}$)**.** *Let $U \subset \Omega$ be a domain and let $S \subset U$ be a discrete set. Assume Θ is holomorphic on $U \setminus S$ and satisfies $|\Theta(s)| \le 1$ for all $s \in U \setminus S$. Then Θ extends holomorphically to U and satisfies $|\Theta| \le 1$ on U. If, moreover, Θ is not identically 1 on any connected component of U, then*

$$2\mathcal{J}(s) \;=\; \frac{1+\Theta(s)}{1-\Theta(s)}$$

extends holomorphically to U with $\Re(2\mathcal{J}) \ge 0$ there; in particular $\mathcal{J}$ has no poles in U.

Proof. Fix $\rho \in S$ and choose a disc $D \subset U$ centered at ρ with $D \cap S = \{\rho\}$. On $D \setminus \{\rho\}$ we have $|\Theta| \le 1$, hence Θ is bounded near ρ and extends holomorphically across ρ by Riemann's theorem (again [8, Ch. 2]). Doing this for each $\rho \in S$ yields a holomorphic extension of Θ to all of U. The bound $|\Theta| \le 1$ persists by continuity.

If $\Theta(s_0) = 1$ at an interior point s_0 of a connected component $U_0 \subset U$, then $|\Theta|$ attains its maximum at s_0, so Θ is constant unimodular on U_0 by the Maximum Modulus Principle [8, Ch. 2]. Thus the hypothesis excludes $\Theta \equiv 1$ on U_0, and therefore $1 - \Theta$ is nonvanishing on U. Hence the Cayley inverse $(1+\Theta)/(1-\Theta)$ is holomorphic on U. Since the Cayley map sends the unit disc into the right half-plane, we have $\Re(2\mathcal{J}) \ge 0$ on U. A holomorphic function cannot have a pole, so $\mathcal{J}$ has no poles in U. □

Conclusion: Schur on a far half-plane implies Theorem 1

By Lemma 4, any zero ρ of ζ in Ω produces a pole of $\mathcal{J}$ in Ω (the numerator factors in (2) are nonzero on Ω).

On $\Omega = \{\Re s > \frac{1}{2}\}$ we have $0 \notin \Omega$, so the compensator $(s-1)/s$ introduces no pole on the working domain. The point $s = 1$ lies in Ω but the factor $(s-1)$ cancels the simple pole of ζ there, so $\mathcal{J}$ is holomorphic at $s = 1$ in the raw gauge $\mathcal{O} \equiv 1$. Whenever a nontrivial gauge $\mathcal{O}$ is introduced, we require (and, for certified bounds, separately verify) that $\mathcal{O}$ is holomorphic and nonvanishing on the stated domain; see Remark 3.

Therefore, if we can certify a Schur bound for Θ on a half-plane $U_\varepsilon = \{\, \Re s > 0.6 - \varepsilon \,\}$ with some $\varepsilon > 0$, then Corollary 6 implies $\mathcal{J}$ has no poles in U_ε, hence ζ has no zeros in U_ε. Since $\{\, \Re s \geq 0.6 \,\} \subset U_\varepsilon$, this yields Theorem 1. The next section discharges the Schur bound by a boundary-certificate route and then specializes to U_ε.

4 All-heights Schur bound via a boundary wedge certificate

We now discharge the Schur bound required in Corollary 6 on a half-plane U_ε. The key input is an unconditional *boundary wedge* (P+) for a suitably outer-normalized version of $\mathcal{J}$ on the boundary line $\Re s = \frac{1}{2}$. This route is analytic (no large-height asymptotics) and applies for all heights.

Outer normalization on $\Re s = \frac{1}{2}$

Define

$$F(s) := \frac{\det_2(I - A(s))}{\zeta(s)} \cdot \frac{s-1}{s}, \qquad (\Re s > \tfrac{1}{2}),$$

and extend F to $\Omega \setminus Z(\zeta)$ by analytic continuation (removing the discrete pole set $Z(\zeta)$).

Lemma 7 (Boundary admissibility and Smirnov class for F)**.** *Let F be as above. Then on each connected component of $\Omega \setminus Z(\zeta)$:*

1. *F belongs to the Smirnov class N^+ and therefore admits nontangential boundary values $F^*(t) = \operatorname{nt-lim}_{\sigma \downarrow \frac{1}{2}} F(\sigma + it)$ for Lebesgue-a.e. $t \in \mathbb{R}$.*

2. *The boundary log-modulus $u(t) := \log |F^*(t)|$ lies in $L^1_{\mathrm{loc}}(\mathbb{R})$.*

Moreover, if $|u(t)| \leq C \log(2 + |t|)$ for $|t| \geq 1$, then $u \in L^1(\mathbb{R}, (1+t^2)^{-1} dt)$.

Proof. These boundary-admissibility inputs are established in Appendix A via the phase–velocity framework and the length-free Whitney/Carleson control: see in particular Appendix Lemma 16 (unsmoothing for the determinant log-modulus) and Lemma 14 (outer reconstruction from a boundary log-modulus). together with the admissible-bump/energy mechanism culminating in Proposition 29 and the Carleson lemmas 17–24. The final weighted-integrability claim is immediate from $\int_{|t|\geq 1} \frac{\log(2+|t|)}{1+t^2}\, dt < \infty$. □

Lemma 8 (Outer factor from boundary modulus on Ω)**.** *Assume Lemma 7 together with $u \in L^1(\mathbb{R}, (1+t^2)^{-1} dt)$. Then there exists a holomorphic function $\mathcal{O}_\zeta$ on Ω, unique up to a unimodular constant, with no zeros on Ω, such that the nontangential boundary values satisfy*

$$\left|\mathcal{O}_\zeta(\tfrac{1}{2} + it)\right| = \left|F^*(t)\right| \qquad \text{for Lebesgue-a.e. } t \in \mathbb{R}.$$

Moreover, $\log |\mathcal{O}_\zeta(s)|$ is the Poisson extension of $u(t)$ from the boundary line $\Re s = \frac{1}{2}$.

Proof. Translate Ω to the right half-plane $\{\,\Re w > 0\,\}$ via $w = s - \frac{1}{2}$. Since $u \in L^1(\mathbb{R}, (1+t^2)^{-1}dt)$, its Poisson extension $U = \mathcal{P}[u]$ is a harmonic function on Ω with nontangential boundary trace u a.e. Choose a harmonic conjugate V of U on Ω and set $\mathcal{O}_\zeta := \exp(U + iV)$. Then $\mathcal{O}_\zeta$ is holomorphic and zero-free on Ω, and by Fatou theory its boundary modulus is $e^{u(t)}$ for a.e. t. Uniqueness up to a unimodular constant is standard. References: Garnett [4, Ch. II], Rosenblum–Rovnyak [5, Ch. 2]. □

Define the outer-normalized ratio

$$\mathcal{J}_{\text{out}}(s) := \frac{F(s)}{\mathcal{O}_\zeta(s)} = \frac{\det_2(I - A(s))}{\mathcal{O}_\zeta(s)\,\zeta(s)} \cdot \frac{s-1}{s}. \tag{4}$$

Then $|\mathcal{J}_{\text{out}}(\frac{1}{2} + it)| = 1$ for Lebesgue-a.e. t.

Boundary wedge (P+)

Let $w(t) := \operatorname{Arg} \mathcal{J}_{\text{out}}(\frac{1}{2} + it)$ be the boundary phase (defined for a.e. t). We say that (P+) holds if there exists $m \in \mathbb{R}$ such that

$$|w(t) - m| < \frac{\pi}{2} \qquad \text{for Lebesgue-a.e. } t \in \mathbb{R}.$$

Equivalently, $\Re\big(e^{-im}\mathcal{J}_{\text{out}}(\frac{1}{2} + it)\big) \geq 0$ for Lebesgue-a.e. t.

Theorem 9 (Boundary wedge (P+))**.** *The boundary wedge* (P+) *holds for* $\mathcal{J}_{\text{out}}$.

Proof. See Appendix A, where we provide a complete proof of the quantitative boundary certificate. □

From (P+) to an interior Schur bound

Fix m witnessing (P+) and set

$$\widetilde{\mathcal{J}}(s) := e^{-im}\mathcal{J}_{\text{out}}(s).$$

Then $|\widetilde{\mathcal{J}}(\frac{1}{2} + it)| = 1$ a.e. and $\Re\,\widetilde{\mathcal{J}}(\frac{1}{2} + it) \geq 0$ a.e. Define the corresponding Cayley field

$$\Theta_{\text{out}}(s) := \frac{2\widetilde{\mathcal{J}}(s) - 1}{2\widetilde{\mathcal{J}}(s) + 1}.$$

Lemma 10 (Smirnov regularity for $\mathcal{J}_{\text{out}}$ and Θ_{out})**.** *Assume Lemmas 7 and 8. Then* $\mathcal{J}_{\text{out}} \in N^+(\Omega \backslash Z(\zeta))$ *and admits nontangential boundary values* $\mathcal{J}_{\text{out}}(\frac{1}{2}+it)$ *for Lebesgue-a.e.* t*. Consequently,* $\widetilde{\mathcal{J}} = e^{-im}\mathcal{J}_{\text{out}} \in N^+(\Omega \setminus Z(\zeta))$*, and* Θ_{out} *admits Lebesgue-a.e. boundary values on* $\Re s = \frac{1}{2}$*.*

Proof. By Lemma 7, $F \in N^+(\Omega \setminus Z(\zeta))$. By Lemma 8, $\mathcal{O}_\zeta$ is holomorphic, zero-free, and outer on Ω, hence lies in $N^+(\Omega)$. Therefore $\mathcal{J}_{\text{out}} = F/\mathcal{O}_\zeta \in N^+(\Omega \setminus Z(\zeta))$. Multiplication by a unimodular constant preserves N^+ membership, so $\widetilde{\mathcal{J}} \in N^+(\Omega \setminus Z(\zeta))$. The boundary-value statements follow from Smirnov boundary theory (e.g. Garnett [4, Ch. II]). Finally, Θ_{out} is a Möbius transform of $\widetilde{\mathcal{J}}$; its a.e. boundary values exist wherever $2\widetilde{\mathcal{J}}(\frac{1}{2} + it) \neq -1$, which holds a.e. since $\Re\,\widetilde{\mathcal{J}}(\frac{1}{2} + it) \geq 0$ a.e. □

Lemma 11 (Boundary-to-interior Schur transport on Ω)**.** *Let* $\Theta \in N^+(\Omega)$ *admit nontangential boundary values* $\Theta(\frac{1}{2} + it)$ *for Lebesgue-a.e.* t*. If* $|\Theta(\frac{1}{2} + it)| \leq 1$ *for Lebesgue-a.e.* t*, then* $|\Theta(s)| \leq 1$ *for all* $s \in \Omega$*.*

Proof. For $\Theta \not\equiv 0$, the function $\log|\Theta|$ is subharmonic on Ω and has a harmonic majorant because $\Theta \in N^+(\Omega)$. At Lebesgue points where the nontangential boundary values exist, the boundary hypothesis gives $\log|\Theta(\frac{1}{2}+it)| \le 0$ a.e. By the Poisson domination principle for subharmonic functions on a half-plane, this implies $\log|\Theta(s)| \le 0$ for all $s \in \Omega$, hence $|\Theta(s)| \le 1$. See Garnett [4, Ch. II], Rosenblum–Rovnyak [5, Ch. 2]. □

Proposition 12 (Herglotz/Schur transport)**.** *Assume* (P+) *for* $\mathcal{J}_{\text{out}}$ *and Lemma 10. Then* Θ_{out} *is Schur on each connected component of* $\Omega \setminus Z(\zeta)$*, and*

$$H(s) := \frac{1+\Theta_{\text{out}}(s)}{1-\Theta_{\text{out}}(s)}$$

is Herglotz on $\Omega \setminus Z(\zeta)$ *(i.e.* $\Re H(s) \ge 0$ *there). Moreover,* $H(s) = 2\widetilde{\mathcal{J}}(s) = 2e^{-im}\mathcal{J}_{\text{out}}(s)$*.*

Proof. (Clarification.) The input from Appendix A used here is: (i) F_ε and hence Θ_{out} admit a.e. nontangential boundary values on $\mathbb{R}$; (ii) the (P+) phase–velocity identity holds in the sense of distributions with respect to Lebesgue-a.e. boundary data; (iii) $\Theta_{\text{out}} \in N^+$ on Ω. All subsequent implications are taken with respect to these a.e. boundary traces (not pointwise boundary values). By (P+), we have $\Re\,\widetilde{\mathcal{J}}(\frac{1}{2}+it) \ge 0$ for Lebesgue-a.e. t. Since the Cayley map $z \mapsto \frac{2z-1}{2z+1}$ sends the closed right half-plane into the closed unit disc, it follows that $|\Theta_{\text{out}}(\frac{1}{2}+it)| \le 1$ for Lebesgue-a.e. t. Applying Lemma 11 on each connected component yields $|\Theta_{\text{out}}(s)| \le 1$ throughout $\Omega \setminus Z(\zeta)$. In particular, Θ_{out} is locally bounded near any point, so any isolated singularity of Θ_{out} is removable. Thus Θ_{out} is holomorphic and bounded by 1 on $\Omega \setminus Z(\zeta)$, i.e. it is Schur there.

Finally, the Cayley inverse maps the unit disc into the closed right half-plane, so $H = (1+\Theta_{\text{out}})/(1-\Theta_{\text{out}})$ is holomorphic on $\Omega \setminus Z(\zeta)$ and satisfies $\Re H \ge 0$ there. A direct algebraic computation shows $H(s) = 2\widetilde{\mathcal{J}}(s) = 2e^{-im}\mathcal{J}_{\text{out}}(s)$. □

Proof of Theorem 1. By Proposition 12, Θ_{out} is Schur on $\Omega \setminus Z(\zeta)$. Apply Corollary 6 to $\widetilde{\mathcal{J}} = e^{-im}\mathcal{J}_{\text{out}}$ on U_ε to conclude that $\widetilde{\mathcal{J}}$ has no poles on U_ε; hence $\mathcal{J}_{\text{out}}$ has no poles on U_ε as well. Since $\det_2(I-A)$ and $\mathcal{O}_\zeta$ are holomorphic and nonvanishing on Ω, poles of $\mathcal{J}_{\text{out}}$ in Ω can only come from zeros of ζ. Therefore ζ has no zeros in U_ε, and in particular none in $\{\,\Re s \ge 0.6\,\}$. □

Table 1: Supplementary computational artifacts (not used in the proof).

Artifact	**Parameter**	**Value**
Rectangle certification (`theta_certify`)		
Domain	$[\sigma_{\min}, \sigma_{\max}] \times [t_{\min}, t_{\max}]$	$[0.6, 0.7] \times [0, 20]$
Certified upper bound	$\max \lvert\Theta_{\text{proj}}\rvert$	0.9999928763
Safety margin	$1 - \theta_{\text{hi}}$	7.12×10^{-6}
Status	`ok`	`true`
Boxes processed		380,764
Precision	(bits)	260
Gauge		`outer_zeta_proj`
Pick certificate (`pick_certify`, $\sigma_0 = 0.599$)		
Matrix size	N	16
Spectral separation	δ_{cert}	0.594
SPD at origin	$P_N \succ 0$	`true`
Coefficient count	N_{coeff}	128
Tail sum (diagnostic)	$\sum_{16}^{127} \lvert a_n\rvert$	0.67
Gauge		`raw_zeta`
Pick certificate (`pick_certify`, $\sigma_0 = 0.6$)		
Matrix size	N	16
Spectral separation	δ_{cert}	0.594
SPD at origin	$P_N \succ 0$	`true`
Coefficient count	N_{coeff}	128
Tail sum (diagnostic)	$\sum_{16}^{127} \lvert a_n\rvert$	0.67
Gauge		`raw_zeta`
Pick certificate (`pick_certify`, $\sigma_0 = 0.7$)		
Matrix size	N	16
Spectral separation	δ_{cert}	0.627
SPD at origin	$P_N \succ 0$	`true`
Coefficient count	N_{coeff}	128
Tail sum (diagnostic)	$\sum_{16}^{127} \lvert a_n\rvert$	0.61
Gauge		`raw_zeta`

Conclusion and limitations (unconditional status)

We prove a fixed half-plane zero-exclusion for the Riemann zeta function: $\zeta(s) \neq 0$ for $\Re s \geq 0.6$ (Theorem 1).

The argument is analytic and function-theoretic: zeros are converted into poles of an arithmetic ratio $\mathcal{J}$, and a Schur bound $|\Theta| \leq 1$ for the associated Cayley field forces removability and rules out poles (hence zeros). For transparency, the handoff bundle also provides independently checkable ball-arithmetic artifacts on representative low-height rectangles together with a verification protocol in Appendix B. The technically hardest step is establishing the all-heights Schur bound, which is discharged by the boundary wedge certificate (P+) (Section 4). The supplementary artifacts in Table 1 are provided for independent verification of the low-height rectangle and Pick cross-checks listed in Appendix B.

Computer assistance and verifiability. Although the core implications are analytic, the repository provides rigorous numerical certificates (ball arithmetic) together with a verifier and JSON outputs so that the finite checks can be independently verified.

Limitations and scope. We do not claim the Riemann Hypothesis here.

All conclusions are confined to the explicit region proved, and no statement is made about zeros with $\Re s < 0.6$. It isolates and certifies a fixed far-field exclusion $\Re s \geq 0.6$. Pushing the boundary 0.6 closer to 1/2 within this framework would require sharpening the analytic boundary-certificate constants and the Carleson/box-energy bounds that enter the wedge criterion, which we do not pursue here. The companion papers in this series treat (i) effective near-field barriers in the strip $1/2 < \Re s < 0.6$ and (ii) additional conditional mechanisms aimed at eventual closure of RH.

Statements and Declarations

Competing interests. The author declares no competing interests.

Data and materials availability. For reproducibility, the verification is with respect to the shipped bundle (and its SHA-256 manifest, if provided). If any file is updated after theory revision, the verification should be rerun and the manifest regenerated. All computational artifacts used for supplementary cross-checks are included in the handoff bundle (and mirrored in the repository):

```
compute/artifacts/theta_certify_sigma06_07_t0_20_outer_zeta_proj.json
compute/artifacts/pick_sigma0599_raw_zeta_N16.json
compute/artifacts/pick_sigma06_raw_zeta_N16.json
compute/artifacts/pick_sigma07_raw_zeta_N16.json
compute/verify_attachment_arb.py
```

Remark 13. The verifier is based on rigorous ball arithmetic (ARB via `python-flint`) and is intended to be independently verifiable. Appendix B records the verification manifest and a minimal command sequence for running the verifier on a fresh machine.

A Proof of the boundary wedge certificate (P+)

This appendix proves Theorem 9. The argument combines a quantitative phase–velocity identity with a Whitney-box Cauchy–Riemann/Green pairing and a Carleson-type energy bound, yielding a windowed phase estimate that implies the boundary wedge inclusion (P+).

A.1 Statement, standing notation, and domains

This subsection fixes the ambient domain, boundary conventions, Whitney geometry, and the meaning of boundary limits, so later phase and energy identities are unambiguous.

Throughout Appendix A we work in the right half-plane

$$\Omega := \{s \in \mathbb{C} : \Re s > \tfrac{1}{2}\},$$

with boundary line $\partial\Omega = \{\frac{1}{2} + it : t \in \mathbb{R}\}$. All analytic objects are understood componentwise on $\Omega \setminus Z$, where Z denotes the relevant zero/pole set, so that branches of log and Arg are well-defined on each connected component.

For a compact interval $I \subset \mathbb{R}$ and a dilation parameter $\alpha > 1$ we write $Q_\alpha(I)$ for the standard Whitney box based on I, and we use the weighted area measure $\sigma\, dt\, d\sigma$ on Ω, where $\sigma := \Re s - \frac{1}{2}$.

The goal is to prove the boundary wedge certificate (P+) stated in Theorem 9. The proof proceeds by: (i) a quantitative phase–velocity identity for the boundary phase of $\mathcal{J}_{\text{out}}$, (ii) a Green/Cauchy–Riemann pairing on Whitney boxes, (iii) a Carleson-type energy bound for a logarithmic derivative, and (iv) a quantitative wedge criterion converting windowed phase control into a.e. wedge inclusion.

In the phase–velocity identity beloww, $-w'$ is a positive distribution (a locally finite measure plus a discrete atomic part) encoding the off–critical zero data that enter the boundary phase derivative.

A.2 A quantitative wedge criterion from Whitney-local control

We state the wedge target (P+) in a form suited to local Whitney-box estimates and record the boundary conventions used throughout Appendix A.

We work on the boundary line $\Re s = \frac{1}{2}$ and use the following conventions.

- *Wedge.* For an aperture parameter $\alpha \in (0, \frac{\pi}{2})$ and a center angle $m \in \mathbb{R}$, write

$$W_{m,\alpha} := \{z \in \mathbb{C} : \ |\operatorname{Arg}(e^{-im}z)| \le \alpha\}.$$

  Thus (P+) is the Lebesgue-a.e. inclusion $\mathcal{J}_{\text{out}}(\frac{1}{2} + it) \in W_{m,\alpha}$ for some fixed $\alpha < \frac{\pi}{2}$ and some $m \in \mathbb{R}$.

- *Whitney / Carleson boxes.* For an interval $I \subset \mathbb{R}$, write the Carleson box $S(I) := \{\frac{1}{2} + \sigma + it : 0 < \sigma \le |I|, \ t \in I\}$. A Whitney box means a box of comparable width and height, e.g. $\{\frac{1}{2} + \sigma + it : \ \sigma \in [a|I|, b|I|], \ t \in I\}$ with fixed $0 < a < b$.

- *Meaning of "a.e."* Unless explicitly stated otherwise, "a.e." refers to Lebesgue measure dt on $\mathbb{R}$.

Lemma 14 (Outer normalizer from boundary log-modulus)**.** *Let $u \in L^1(\mathbb{R}, (1+t^2)^{-1}dt)$ be real-valued. Then there exists an outer function O on Ω (zero-free and holomorphic on Ω) whose nontangential boundary values satisfy*

$$|O(\tfrac{1}{2} + it)| = e^{u(t)} \quad \textit{for a.e. } t \in \mathbb{R}.$$

Moreover O is unique up to a unimodular constant.

Proof. Define the Poisson extension U of u to Ω by

$$U(\tfrac{1}{2} + \sigma + it) \ := \ \frac{1}{\pi} \int_{\mathbb{R}} u(\tau) \, \frac{\sigma}{\sigma^2 + (t-\tau)^2} \, d\tau, \qquad \sigma > 0.$$

The weighted integrability $u \in L^1(\mathbb{R}, (1+t^2)^{-1}dt)$ ensures the integral converges and that U is harmonic on Ω. Let V be a harmonic conjugate of U on Ω (defined up to an additive constant), and set

$$O(s) \ := \ \exp\big(U(s) + iV(s)\big).$$

Then O is holomorphic and zero-free on Ω. Standard boundary theory for Poisson integrals gives that $U(\frac{1}{2} + \varepsilon + it) \to u(t)$ for a.e. t as $\varepsilon \downarrow 0$; therefore the nontangential boundary values satisfy $|O(\frac{1}{2} + it)| = e^{u(t)}$ for a.e. t. Uniqueness up to unimodular constant follows because the ratio of two such outer functions has a.e. boundary modulus 1 and hence is an inner constant. (Source.) This is the classical outer-function construction in half-planes; see Duren H^p *Spaces*, Ch. II, or Garnett *Bounded Analytic Functions*, Ch. II. □

A.3 Phase–velocity identity (quantitative form) and boundary passage

We establish the boundary phase–velocity relation for the outer-normalized ratio, and record the precise sense in which derivatives and boundary traces are taken.

Lemma 15 (Outer–Hilbert boundary identity)**.** *Let $u \in L^1_{\mathrm{loc}}(\mathbb{R})$ and let O be an outer function on Ω whose boundary modulus satisfies $|O(\frac{1}{2}+it)| = e^{u(t)}$ for a.e. t. Let $w(t) := \operatorname{Arg} O(\frac{1}{2}+it)$ denote the boundary argument (defined modulo an additive constant). Then, in $\mathcal{D}'(\mathbb{R})$,*

$$\frac{d}{dt} w(t) = \mathcal{H}[u'](t),$$

where $\mathcal{H}$ is the boundary Hilbert transform on $\mathbb{R}$ (as a continuous operator $\mathcal{D}'(\mathbb{R}) \to \mathcal{D}'(\mathbb{R})$) and u' is the distributional derivative.

Proof. Write $\log O = U + iV$ on Ω, where $U = \Re \log O$ is harmonic and $V = \Im \log O$ is its harmonic conjugate (on each component of $\Omega \setminus Z(O)$, fixing a branch of log). The boundary trace satisfies $U(\frac{1}{2}+\cdot) = u$ in $\mathcal{D}'(\mathbb{R})$, and the conjugate boundary trace is $V(\frac{1}{2}+\cdot) = \mathcal{H}[u]$ in $\mathcal{D}'(\mathbb{R})$ (up to an additive constant). Differentiating in t in the sense of distributions gives

$$\frac{d}{dt} \operatorname{Arg} O(\tfrac{1}{2}+it) = \partial_t V(\tfrac{1}{2}+it) = \mathcal{H}[\partial_t u](t) = \mathcal{H}[u'](t),$$

since differentiation commutes with $\mathcal{H}$ on $\mathcal{D}'(\mathbb{R})$. □

Lemma 16 (Smoothed distributional bound for $\partial_\sigma \Re \log \det_2$)**.** *Let $I \Subset \mathbb{R}$ be a compact interval and fix $\varepsilon_0 \in (0, \frac{1}{2}]$. There exists a finite constant*

$$C_* \;:=\; \sum_p \sum_{k\ge 2} \frac{p^{-k/2}}{k^2 \,\log p} \;<\; \infty$$

such that for all $\sigma \in (\frac{1}{2}, \frac{1}{2}+\varepsilon_0]$ and every $\varphi \in C_c^2(I)$,

$$\Big| \int_{\mathbb{R}} \varphi(t)\, \partial_\sigma \Re \log \det_2 \big(I - A(\sigma+it)\big)\, dt \Big| \;\le\; C_* \, \|\varphi''\|_{L^1(I)}.$$

Proof. For $\sigma > \frac{1}{2}$ one has the absolutely convergent expansion

$$\partial_\sigma \Re \log \det_2 \big(I - A(\sigma+it)\big) \;=\; \sum_p \sum_{k\ge 2} (\log p)\, p^{-k\sigma} \cos(kt \log p).$$

For each frequency $\omega = k \log p \ge 2 \log 2$, two integrations by parts give

$$\Big| \int_{\mathbb{R}} \varphi(t) \cos(\omega t)\, dt \Big| \;\le\; \frac{\|\varphi''\|_{L^1(I)}}{\omega^2}.$$

Summing the resulting majorant yields

$$\Big| \int \varphi\, \partial_\sigma \Re \log \det_2 \, dt \Big| \;\le\; \|\varphi''\|_{L^1} \sum_p \sum_{k\ge 2} \frac{(\log p)\, p^{-k\sigma}}{(k \log p)^2} \;\le\; \|\varphi''\|_{L^1} \sum_p \sum_{k\ge 2} \frac{p^{-k/2}}{k^2 \,\log p},$$

uniformly for $\sigma \in (\frac{1}{2}, \frac{1}{2}+\varepsilon_0]$, since the rightmost double series converges. □

Lemma 17 (Arithmetic Carleson energy)**.** *Let*

$$U_{\det_2}(\sigma,t) \;:=\; \Re\log\det_2\Big(I - A\big(\tfrac{1}{2}+\sigma+it\big)\Big) \;=\; -\sum_p\sum_{k\ge 2}\frac{p^{-k/2}}{k}\,e^{-k\log p\,\sigma}\,\cos\big(k\log p\,t\big), \qquad \sigma>0,$$

where the series converges absolutely for every $\sigma>0$. Then for every interval $I\subset\mathbb{R}$ with Carleson box $Q(I):=I\times(0,|I|]$,

$$\iint_{Q(I)} |\nabla U_{\det_2}|^2\,\sigma\,dt\,d\sigma \;\le\; \frac{|I|}{4}\sum_p\sum_{k\ge 2}\frac{p^{-k}}{k^2} \;=:\; K_0\,|I|, \qquad K_0:=\frac{1}{4}\sum_p\sum_{k\ge 2}\frac{p^{-k}}{k^2}<\infty.$$

Proof. For a single mode $b\,e^{-\omega\sigma}\cos(\omega t)$ one has $|\nabla|^2 = b^2\omega^2 e^{-2\omega\sigma}$, hence

$$\int_0^{|I|}\int_I |\nabla|^2\,\sigma\,dt\,d\sigma \;\le\; |I|\cdot\sup_{\omega>0}\int_0^{|I|}\sigma\,\omega^2 e^{-2\omega\sigma}d\sigma\cdot b^2 \;\le\; \tfrac{1}{4}\,|I|\,b^2.$$

With $b=p^{-k/2}/k$ and $\omega=k\log p$, summing over (p,k) gives the claim and the finiteness of K_0. □

Whitney scale and short–interval zero counts. Throughout the boundary-certificate route we work on Whitney boxes based at height T with

$$L=L(T):=\min\Big\{\frac{c}{\log\langle T\rangle},\; L_\star\Big\}, \qquad \langle T\rangle:=\sqrt{1+T^2}, \qquad c\in(0,1] \text{ fixed.}$$

The only input about the *number* of zeros used beloww is the classical short-interval consequence of Riemann–von Mangoldt: there exist absolute constants $A_0,A_1>0$ such that for $T\ge 2$ and $0<H\le 1$,

$$N(T;H) \;:=\; \#\{\rho=\beta+i\gamma:\;\gamma\in[T,T+H]\} \;\le\; A_0 \;+\; A_1\,H\,\log\langle T\rangle.$$

Lemma 18 (Annular Poisson–balayage L^2 bound)**.** *Let $I=[T-L,T+L]$, $Q_\alpha(I)=I\times(0,\alpha L]$, and fix $k\ge 1$. For $\mathcal{A}_k:=\{\rho=\beta+i\gamma:\;2^kL<|T-\gamma|\le 2^{k+1}L\}$ set*

$$V_k(\sigma,t):=\sum_{\rho\in\mathcal{A}_k}\frac{\sigma}{(t-\gamma)^2+\sigma^2}.$$

Then

$$\iint_{Q_\alpha(I)} V_k(\sigma,t)^2\,\sigma\,dt\,d\sigma \;\ll_\alpha\; |I|\,4^{-k}\,\nu_k,$$

where $\nu_k:=\#\mathcal{A}_k$, and the implicit constant depends only on α.

Proof. Write $K_\sigma(x):=\sigma/(x^2+\sigma^2)$ and $V_k=\sum_{\rho\in\mathcal{A}_k}K_\sigma(\cdot-\gamma)$. Integrate over $t\in I$ first. For the diagonal terms, using $|t-\gamma|\ge 2^kL-L\ge 2^{k-1}L$ for $t\in I$ and $k\ge 1$,

$$\int_I K_\sigma(t-\gamma)^2\,dt=\sigma^2\int_I\frac{dt}{\big((t-\gamma)^2+\sigma^2\big)^2} \;\le\; \frac{L}{(2^{k-1}L)^2}\,\sigma.$$

Multiplying by the area weight σ and integrating $\sigma\in(0,\alpha L]$ gives a contribution $\ll_\alpha |I|\,4^{-k}$ per γ, hence $\ll_\alpha |I|\,4^{-k}\nu_k$ after summing. For off-diagonal terms, for $i\ne j$ one has on I that $K_\sigma(t-\gamma_j)\le\sigma/(2^{k-1}L)^2$, hence

$$\int_I K_\sigma(t-\gamma_i)K_\sigma(t-\gamma_j)\,dt \;\le\; \frac{\sigma}{(2^{k-1}L)^2}\int_{\mathbb{R}}K_\sigma(t-\gamma_i)\,dt=\frac{\pi\sigma}{(2^{k-1}L)^2},$$

and integrating $\sigma\in(0,\alpha L]$ with the extra factor σ yields $\ll_\alpha |I|\,4^{-k}$. Summing over pairs (i,j) via a Schur test gives the stated bound (absorbing constants into $\ll_\alpha$). □

A.4 Quantitative phase–velocity identity

This subsection derives the quantitative identity linking the distribution $-w'$ to the off-axis zero data, in a form usable under Whitney localization.

Lemma 19 (Distributional phase–velocity identity for outer data)**.** *Let $U^* \in L^1_{\mathrm{loc}}(\mathbb{R})$ and define $w := \mathcal{H}[U^*] \in \mathcal{D}'(\mathbb{R})$ by the duality*

$$-\langle w, \varphi' \rangle \;=\; \int_{\mathbb{R}} U^*(t)\,(\mathcal{H}\varphi)'(t)\,dt \qquad \forall\, \varphi \in C_c^\infty(\mathbb{R}).$$

Then $w' = \mathcal{H}[(U^*)']$ in $\mathcal{D}'(\mathbb{R})$.

Proof. Let η_ε be a standard mollifier and set $U^*_\varepsilon := U^* * \eta_\varepsilon \in C^\infty(\mathbb{R})$. For smooth data one has $(\mathcal{H}[U^*_\varepsilon])' = \mathcal{H}[(U^*_\varepsilon)']$ pointwise. Since $U^*_\varepsilon \to U^*$ in L^1_{loc}, we have $(U^*_\varepsilon)' \to (U^*)'$ in $\mathcal{D}'$, and the Hilbert transform is continuous on $\mathcal{D}'$. Passing to the limit yields $w' = \mathcal{H}[(U^*)']$ in $\mathcal{D}'$. □

Lemma 20 (Local L^1 control for $\log|\xi|$ along vertical approach)**.** *Fix a compact interval $I \Subset \mathbb{R}$. Then the family $t \mapsto \log|\xi(\frac{1}{2}+\varepsilon+it)|$ is bounded in $L^1(I)$ uniformly for $\varepsilon \in (0,1]$. Moreover, for $\varepsilon, \varepsilon' \downarrow 0$ the difference $\log|\xi(\frac{1}{2}+\varepsilon+it)| - \log|\xi(\frac{1}{2}+\varepsilon'+it)|$ tends to 0 in $L^1(I)$.*

Proof. This is a standard boundary-approach property for $\log|\xi|$ in the half-plane, obtained from the Poisson representation of $\log|\xi|$ together with classical zero-counting bounds for ξ (equivalently for ζ) on vertical lines; see, e.g., [3, Ch. 9]. □

Theorem 21 (Quantified phase–velocity identity and boundary passage)**.** *Let*

$$u_\varepsilon(t) := \log\big|\det{}_2(I - A(\tfrac{1}{2}+\varepsilon+it))\big| - \log\big|\xi(\tfrac{1}{2}+\varepsilon+it)\big|.$$

Then u_ε is uniformly L^1-bounded and Cauchy on every compact $I \Subset \mathbb{R}$ as $\varepsilon \downarrow 0$, hence $u_\varepsilon \to u$ in $L^1_{\mathrm{loc}}(\mathbb{R})$. Let $\mathcal{O}$ be the outer function on Ω with boundary modulus e^u and normalization $\mathcal{O}(\frac{3}{2}) > 0$, and set

$$\mathcal{J}(s) := \frac{\det_2(I - A(s))}{\mathcal{O}(s)\,\xi(s)}.$$

Then $|\mathcal{J}(\frac{1}{2}+it)| = 1$ for a.e. $t \in \mathbb{R}$. By Lemma 22, $\mathcal{J}$ is of bounded type on every Whitney region $Q_\alpha(I)$. Let $w \in \mathcal{D}'(\mathbb{R})$ denote the distributional boundary phase of $\mathcal{J}$ (defined modulo an additive constant).

Lemma 22 (Local bounded-type control for $\mathcal{J}$)**.** *Fix a compact interval $I \Subset \mathbb{R}$. Assume that $\det_2(I - A(s))$ is holomorphic and nonvanishing on a neighborhood of the Whitney region $Q_\alpha(I) \Subset \Omega$, and that the Carleson energy bounds of Lemmas 17 and 24 hold on $Q_\alpha(I)$. Then $\mathcal{J}$ belongs to the Nevanlinna class (bounded type) on $Q_\alpha(I)$.*

Proof. Write $\mathcal{J} = \det_2(I - A)/(\mathcal{O}\,\xi)$ as above. On each simply connected subdomain of $Q_\alpha(I)$ avoiding zeros, choose branches of $\log\det_2(I - A)$ and $\log\xi$ and set $U := \Re\log\det_2(I - A)$ and $V := \Re\log\xi$. By Lemmas 17 and 24, the measures $|\nabla U|^2\,\sigma\,dt\,d\sigma$ and $|\nabla V|^2\,\sigma\,dt\,d\sigma$ are Carleson on $Q_\alpha(I)$ (with $\sigma = \Re s - \frac{1}{2}$). Standard half-plane theory (Carleson energy $\Rightarrow$ BMO boundary trace $\Rightarrow$ BMOA logarithm) implies that $\det_2(I - A)$ and ξ are of bounded type on $Q_\alpha(I)$; see, e.g., Garnett, *Bounded Analytic Functions*, Ch. VI, and Koosis, *The Logarithmic Integral*. The outer function $\mathcal{O}$ with boundary modulus e^u is an outer Smirnov function on $Q_\alpha(I)$ and hence of bounded type there. Since the Nevanlinna class is closed under products and quotients (where defined), $\mathcal{J}$ is of bounded type on $Q_\alpha(I)$. □

Then, for every compact interval $I \Subset \mathbb{R}$ and every nonnegative $\phi \in C_c^\infty(I)$,

$$\int_I \phi(t)\,(-w'(t))\,dt \;=\; \pi\int_I \phi(t)\,d\mu_{\mathrm{off}}(t) \;+\; \pi\int_I \phi(t)\,d\nu_{\mathrm{sing}}(t) \;+\; \pi\sum_{\gamma\in I} m_\gamma\,\phi(\gamma), \tag{5}$$

where:

- *μ_{off} is the Poisson balayage of the off–critical zeros $\rho = \beta + i\gamma$ of ζ with $\beta > \frac{1}{2}$, counted with multiplicity m_ρ;*
- *ν_{sing} is the (possibly zero) singular boundary measure of any singular inner factor in the canonical factorization of $\mathcal{J}$ on Ω; and*
- *the discrete sum ranges over boundary zeros/poles on $\Re s = \frac{1}{2}$, written $s = \frac{1}{2} + i\gamma$, with multiplicities m_γ.*

Proof. The L^1_{loc} convergence $u_\varepsilon \to u$ is as stated. The outer function $\mathcal{O}$ exists by Lemma 14.

Under the bounded-type hypothesis, $\mathcal{J}$ admits the canonical half-plane factorization into a unimodular constant, a Blaschke product over zeros in Ω, a (possibly trivial) singular inner factor, and an outer factor. Since $|\mathcal{J}(\frac{1}{2} + it)| = 1$ a.e., the outer factor is unimodular constant. Taking the distributional boundary argument w and differentiating in $\mathcal{D}'$, each factor contributes additively: the Blaschke product yields the Poisson balayage measure μ_{off}, the singular inner factor yields ν_{sing}, and boundary zeros/poles yield atomic Dirac masses. This is the standard phase-derivative computation for bounded-type functions on a half-plane; see, e.g., Garnett *Bounded Analytic Functions*, Ch. II, or Koosis *The Logarithmic Integral*, Vol. I. □

A.5 Final assembly ingredients for the boundary wedge certificate (P+)

This subsection collects the three final analytic ingredients—Poisson plateau, Whitney box pairing, and Carleson-energy control—used to conclude (P+).

Poisson plateau lower bound. We prove the key lower bound for the Poisson kernel averaged over admissible windows, which converts discrete off-axis contributions into a uniform wedge aperture.

Lemma 23 (Poisson plateau lower bound)**.** *Let $\psi \in C_c^\infty(\mathbb{R})$ be even with $\psi \equiv 1$ on $[-1,1]$ and* $\operatorname{supp}\psi \subset [-2,2]$. *Then*

$$c_0(\psi) \;:=\; \inf_{0<b\le 1,\ |x|\le 1} (P_b * \psi)(x) \;\ge\; \frac{1}{2\pi}\arctan 2 \;>\; 0.$$

Proof. Since $\psi \ge \mathbf{1}_{[-1,1]}$, it suffices to compute $(P_b * \mathbf{1}_{[-1,1]})(x)$. For $|x| \le 1$,

$$(P_b * \mathbf{1}_{[-1,1]})(x) = \frac{1}{\pi}\int_{-1}^{1} \frac{b}{b^2 + (x-y)^2}\,dy = \frac{1}{2\pi}\Big(\arctan\frac{1-x}{b} + \arctan\frac{1+x}{b}\Big).$$

This expression is minimized over $0 < b \le 1$, $|x| \le 1$, at $(x,b) = (1,1)$, yielding $\frac{1}{2\pi}\arctan 2$. □

From phase–velocity and CR–Green to (P+). We combine the phase–velocity identity with a Cauchy–Riemann/Green pairing on Whitney boxes to obtain the windowed phase control that underlies (P+).

Carleson energy bound for the logarithmic derivative. We prove the weighted L^2 (Carleson-energy) estimate for the relevant logarithmic derivative on Whitney boxes, including the neutralization of near-field zeros.

Lemma 24 (Analytic (ξ) Carleson energy on Whitney boxes)**.** *There exist absolute constants $c \in (0,1]$ and $C_\xi < \infty$ such that for every interval $I = [T-L,\, T+L]$ at Whitney scale $L = c/\log\langle T\rangle$, the Poisson extension*

$$U_\xi(\sigma,t) := \Re \log \xi\big(\tfrac{1}{2}+\sigma+it\big) \qquad (\sigma > 0)$$

obeys the Carleson bound

$$\iint_{Q(I)} |\nabla U_\xi(\sigma,t)|^2\, \sigma\, dt\, d\sigma \;\le\; C_\xi\, |I|.$$

Proof. Fix $I = [T-L, T+L]$ with $L = c/\log\langle T\rangle$ and a fixed aperture $\alpha \in [1,2]$. Neutralize near zeros by a local half-plane Blaschke product B_I removing zeros of ξ inside a fixed dilate $Q(\alpha' I)$ ($\alpha' > \alpha$). This yields a harmonic field $\widetilde{U}_\xi$ on $Q(\alpha I)$ and

$$\iint_{Q(\alpha I)} |\nabla U_\xi|^2\, \sigma\, dt\, d\sigma \;\asymp\; \iint_{Q(\alpha I)} |\nabla \widetilde{U}_\xi|^2\, \sigma\, dt\, d\sigma \;+\; O_\alpha(|I|),$$

so it suffices to bound the neutralized energy.

Choose a branch of $\log\xi$ on each component of $Q(\alpha I)\setminus Z(\xi)$ so that $U_\xi = \Re\log\xi$ is harmonic there. Then

$$|\nabla U_\xi(\sigma,t)|^2 \;=\; |(\log\xi)'(\tfrac{1}{2}+\sigma+it)|^2 \;=\; \Big|\frac{\xi'}{\xi}(\tfrac{1}{2}+\sigma+it)\Big|^2.$$

By the Hadamard product for ξ, $\xi'/\xi(s) = A(s) + \sum_\rho (s-\rho)^{-1}$, where A is holomorphic and slowly varying on compact strips. Thus it suffices to bound the weighted L^2-norm of ξ'/ξ on $Q(\alpha I)$; the contribution of A is $O_\alpha(|I|)$, and the remaining term is handled annularly by summing the zero contributions. Decompose the (neutralized) zeros into Whitney annuli $\mathcal{A}_k := \{\rho : 2^k L < |\gamma - T| \le 2^{k+1}L\}$, $k \ge 1$. For $k \ge 1$ and $t \in I$, any $\rho = \beta + i\gamma \in \mathcal{A}_k$ satisfies $|t-\gamma| \ge 2^k L - L \ge 2^{k-1}L$. Since $|s-\rho|^2 = (t-\gamma)^2 + (\frac{1}{2}+\sigma-\beta)^2 \ge (t-\gamma)^2$, we have the pointwise bound $|(s-\rho)^{-1}| \le |t-\gamma|^{-1} \le (2^{k-1}L)^{-1}$ for all $s = \frac{1}{2}+\sigma+it \in Q_\alpha(I)$. Therefore, writing $S_k(s) := \sum_{\rho\in\mathcal{A}_k}(s-\rho)^{-1}$,

$$|S_k(s)|^2 \;\le\; \nu_k \sum_{\rho\in\mathcal{A}_k} |s-\rho|^{-2} \;\le\; \nu_k \cdot \nu_k\, (2^{k-1}L)^{-2} \;=\; \nu_k^2\, (2^{k-1}L)^{-2}.$$

Integrating this uniform bound over $Q_\alpha(I)$ with the area weight $\sigma\, dt\, d\sigma$ gives

$$\iint_{Q_\alpha(I)} |S_k(s)|^2\, \sigma\, dt\, d\sigma \;\le\; \nu_k^2\, (2^{k-1}L)^{-2} \cdot |I| \cdot \int_0^{\alpha L} \sigma\, d\sigma \;\ll_\alpha\; |I|\, L^2\, \frac{\nu_k^2}{4^k\, L^2} \;=\ll_\alpha\; |I|\, 4^{-k}\, \nu_k^2.$$

Now sum over annuli using Cauchy–Schwarz in k: $\big|\sum_{\rho\notin Q(\alpha' I)}(s-\rho)^{-1}\big|^2 \;\le\; (\sum_k |S_k(s)|)^2 \;\le\; (\sum_k 2^{-k})\,(\sum_k 2^k |S_k(s)|^2)$, hence after integrating,

$$\iint_{Q_\alpha(I)} \Big|\sum_{\rho\notin Q(\alpha' I)} (s-\rho)^{-1}\Big|^2\, \sigma\, dt\, d\sigma \;\ll\; \sum_{k\ge1} 2^k \iint_{Q_\alpha(I)} |S_k(s)|^2\, \sigma\, dt\, d\sigma \;\ll_\alpha\; |I| \sum_{k\ge1} 2^{-k}\, \nu_k^2.$$

To bound ν_k, use the short-interval zero-count bound above to obtain, for some absolute $a_1(\alpha), a_2(\alpha)$,

$$\nu_k \;\le\; a_1(\alpha)\, 2^k L\, \log\langle T\rangle \;+\; a_2(\alpha)\, \log\langle T\rangle.$$

Therefore, using $\nu_k \ll_\alpha 2^k L \log\langle T\rangle + \log\langle T\rangle$, we obtain

$$\sum_{k\ge 1} 2^{-k}\,\nu_k^2 \;\ll\; \sum_{k\ge 1} 2^{-k}\big(4^k L^2 \log^2\langle T\rangle + \log^2\langle T\rangle\big) \;\ll\; L^2\log^2\langle T\rangle + \log^2\langle T\rangle.$$

On Whitney scale $L = c/\log\langle T\rangle$, this is $\ll 1$. Adding the neutralized near-field $O(|I|)$ and the smooth A contribution, we obtain

$$\iint_{Q(\alpha I)} |\nabla U_\xi|^2\,\sigma\,dt\,d\sigma \;\le\; C_\xi\,|I|,$$

with C_ξ depending only on (α, c). □

A.6 From windowed phase control to the wedge

We convert the established windowed phase bounds into an almost-everywhere wedge inclusion for the boundary values of $\mathcal{J}_{\text{out}}$.

Definition 25 (Admissible window class with atom avoidance)**.** Fix an even C^∞ window ψ with $\psi \equiv 1$ on $[-1,1]$ and $\operatorname{supp}\psi \subset [-2,2]$. For an interval $I = [t_0 - L, t_0 + L]$, an aperture $\alpha' > 1$, and a parameter $\varepsilon \in (0, \frac{1}{4}]$, define $\mathcal{W}_{\text{adm}}(I;\varepsilon)$ to be the set of C^∞, nonnegative, mass-1 bumps ϕ supported in the fixed dilate $2I = [t_0 - 2L, t_0 + 2L]$ that can be written as

$$\phi(t) \;=\; \frac{1}{Z}\frac{1}{L}\,\psi\!\left(\frac{t-t_0}{L}\right) m(t), \qquad Z = \int_{2I} \frac{1}{L}\psi\!\left(\frac{t-t_0}{L}\right) m(t)\,dt,$$

where $2I := [t_0 - 2L, t_0 + 2L]$ and the mask $m \in C^\infty(2I;[0,1])$ satisfies:

(i) *Atom avoidance.* There is a union of disjoint open subintervals $E = \bigcup_{j=1}^{J} J_j \subset I$ with total length $|E| \le \varepsilon L$ such that $m \equiv 0$ on E and $m \equiv 1$ on $I \setminus E'$, where each transition layer $E' \setminus E$ has thickness $\le \varepsilon L$.

(ii) *Uniform smoothness.* $\|m'\|_\infty \lesssim (\varepsilon L)^{-1}$ and $\|m''\|_\infty \lesssim (\varepsilon L)^{-2}$ with implicit constants independent of I, t_0, L and of the number/placement of the holes $\{J_j\}$.

Every $\phi \in \mathcal{W}_{\text{adm}}(I;\varepsilon)$ is supported in $2I$. This class contains the unmasked profile $\varphi_{L,t_0}(t) = Z_0^{-1}L^{-1}\psi((t-t_0)/L)$ with $Z_0 := \int_{-2}^{2}\psi(x)\,dx$ (take $E = \varnothing$, $m \equiv 1$) and also allows dodging boundary atoms by punching out small neighborhoods while keeping total deleted length $\le \varepsilon L$.

Lemma 26 (Uniform Poisson–energy bound for admissible tests)**.** *Let V_ϕ be the Poisson extension of $\phi \in \mathcal{W}_{\text{adm}}(I;\varepsilon)$ to the half-plane, and fix a cutoff to $Q(\alpha' I)$ with $\alpha' > 1$ as in the CR–Green pairing. Then there exists a finite constant $\mathcal{A}_{\text{adm}}(\psi,\varepsilon,\alpha') < \infty$, depending only on $(\psi,\varepsilon,\alpha')$, such that*

$$\iint_{Q(\alpha' I)} |\nabla V_\phi(\sigma,t)|^2\,\sigma\,dt\,d\sigma \;\le\; \mathcal{A}_{\text{adm}}(\psi,\varepsilon,\alpha')^2\,L.$$

Proof. Let $\phi(t) = Z^{-1}L^{-1}\psi((t-t_0)/L)m(t)$ be an admissible test. By scaling of the Poisson kernel and the uniform bounds on m, m', m'' from Definition 25, the H^1-size of ϕ (equivalently the $L^2(\sigma)$ Dirichlet energy of its Poisson extension on a fixed aperture box) is controlled uniformly by a constant depending only on $(\psi,\varepsilon,\alpha')$, times $L^{1/2}$. Squaring yields the stated $\lesssim L$ energy bound with $\mathcal{A}_{\text{adm}}(\psi,\varepsilon,\alpha')$. □

Lemma 27 (Cutoff pairing on boxes)**.** *Fix parameters $\alpha' > \alpha > 1$. Let $\chi_{L,t_0} \in C_c^\infty(\mathbb{R}^2_+)$ satisfy $\chi \equiv 1$ on $Q(\alpha I)$, $\operatorname{supp}\chi \subset Q(\alpha' I)$, $\|\nabla\chi\|_\infty \lesssim L^{-1}$ and $\|\nabla^2\chi\|_\infty \lesssim L^{-2}$. Let V_ϕ be the Poisson extension of $\phi \in \mathcal{W}_{\mathrm{adm}}(I;\varepsilon)$. Then one has the Green pairing identity*

$$\int_{\mathbb{R}} u(t)\,\phi(t)\,dt \;=\; \iint_{Q(\alpha' I)} \nabla U \cdot \nabla\big(\chi_{L,t_0}\,V_\phi\big)\,dt\,d\sigma \;+\; \mathcal{R}_{\mathrm{side}} \;+\; \mathcal{R}_{\mathrm{top}},$$

with remainders satisfying

$$|\mathcal{R}_{\mathrm{side}}| + |\mathcal{R}_{\mathrm{top}}| \;\lesssim\; \Big(\iint_{Q(\alpha' I)} |\nabla U|^2\,\sigma\Big)^{1/2} \cdot \Big(\iint_{Q(\alpha' I)} \big(|\nabla\chi|^2\,|V_\phi|^2 + |\nabla V_\phi|^2\big)\,\sigma\Big)^{1/2}.$$

Proof. Let $Q := Q(\alpha' I)$. Assume U is C^2 on $\overline{Q}$ and harmonic on Q, with boundary trace $u(t) = U(0,t)$ on the bottom edge $\{\sigma = 0\}$. Since $\chi_{L,t_0}V_\phi$ is compactly supported in $\overline{Q}$ and smooth on Q, Green's identity gives

$$\iint_Q \nabla U \cdot \nabla(\chi V_\phi)\,dt\,d\sigma \;=\; \int_{\partial Q} (\chi V_\phi)\,\partial_n U\,ds \;-\; \iint_Q (\chi V_\phi)\,\Delta U\,dt\,d\sigma.$$

Since $\Delta U = 0$ on Q, only the boundary integral remains. On the bottom edge one has $\partial_n = -\partial_\sigma$, $\chi \equiv 1$, and $V_\phi(0,t) = \phi(t)$, hence that contribution equals

$$\int_I \phi(t)\,(-\partial_\sigma U)(0,t)\,dt.$$

A.7 Whitney box pairing and local oscillation functional

We organize the remaining bookkeeping: pairing Whitney boxes across scales and bounding the oscillation functional needed to pass from local control to the global wedge statement.

If U is the real part of a holomorphic logarithm $U = \Re\log J$ with $|J(\frac{1}{2}+it)| = 1$ a.e., then $U(0,t) = 0$ a.e. and $-\partial_\sigma U(0,t) = \partial_t \operatorname{Arg} J(\frac{1}{2}+it)$ in distributions by Cauchy–Riemann; in particular, this term is the tested boundary phase derivative in Lemma 28 beloww. The remaining boundary pieces (two vertical sides and the top edge) are, by definition, the remainders $\mathcal{R}_{\mathrm{side}} + \mathcal{R}_{\mathrm{top}}$.

For the remainder estimate, we apply Cauchy–Schwarz in the scale-invariant measure $\sigma\,dt\,d\sigma$ on Q:

$$|\mathcal{R}_{\mathrm{side}}| + |\mathcal{R}_{\mathrm{top}}| \;\lesssim\; \Big(\iint_Q |\nabla U|^2\,\sigma\Big)^{1/2}\Big(\iint_Q \big|\nabla(\chi V_\phi)\big|^2\,\sigma\Big)^{1/2}.$$

Expanding $\nabla(\chi V_\phi) = \chi\,\nabla V_\phi + (\nabla\chi)\,V_\phi$ yields

$$\iint_Q \big|\nabla(\chi V_\phi)\big|^2\,\sigma \;\lesssim\; \iint_Q \big(|\nabla V_\phi|^2 + |\nabla\chi|^2|V_\phi|^2\big)\,\sigma,$$

which gives the displayed estimate. □

Lemma 28 (CR–Green pairing for boundary phase)**.** *Let J be analytic on Ω with a.e. boundary modulus $|J(\frac{1}{2}+it)| = 1$, and write $\log J = U + iW$ on Ω, so U is harmonic with $U(\frac{1}{2}+it) = 0$ a.e. Fix a Whitney interval $I = [t_0 - L, t_0 + L]$ and let V_ϕ be the Poisson extension of $\phi \in \mathcal{W}_{\mathrm{adm}}(I;\varepsilon)$. Then, with a cutoff χ_{L,t_0} as in Lemma 27,*

$$\int_{\mathbb{R}} \phi(t)\,\big(-W'(t)\big)\,dt \;=\; \iint_{Q(\alpha' I)} \nabla U \cdot \nabla\big(\chi_{L,t_0}\,V_\phi\big)\,dt\,d\sigma \;+\; \mathcal{R}_{\mathrm{side}} \;+\; \mathcal{R}_{\mathrm{top}},$$

and the remainders satisfy the same estimate as in Lemma 27. In particular, by Cauchy–Schwarz and Lemma 26, there is a constant $C_{\mathrm{rem}}(\alpha',\psi)$ such that

$$\int_{\mathbb{R}} \phi(t)\,\big(-w'(t)\big)\,dt \;\le\; C_{\mathrm{rem}}(\alpha',\psi)\,\Big(\iint_{Q(\alpha' I)} |\nabla U|^2\,\sigma\Big)^{1/2}.$$

Proof. On the bottom edge $\{\sigma = 0\}$ the outward normal is $\partial_n = -\partial_\sigma$. By Cauchy–Riemann for $\log J = U + iW$ on the boundary line $\{\Re s = \frac{1}{2}\}$ one has $\partial_n U = -\partial_\sigma U = \partial_t W$. Thus the bottom-edge term in Green's identity is

$$-\int_{\partial Q \cap \{\sigma=0\}} \chi\, V_\phi\, \partial_n U\, dt = -\int_{\mathbb{R}} \phi(t)\, \partial_t W(t)\, dt = \int_{\mathbb{R}} \phi(t)\, \big(-w'(t)\big)\, dt,$$

which yields the stated identity after including the interior term and remainders. The final inequality is Cauchy–Schwarz together with the uniform Poisson-energy bound from Lemma 26. □

Proposition 29 (Length-independent upper bound for admissible tests)**.** *Let J be holomorphic on $\Omega \setminus Z(\zeta)$ with a.e. boundary modulus 1, write $\log J = U + iW$ on $\Omega \setminus Z(\zeta)$, and let $-w'$ denote the boundary phase distribution. For every interval $I = [t_0 - L, t_0 + L]$, every $\phi \in \mathcal{W}_{\mathrm{adm}}(I;\varepsilon)$, and every fixed cutoff to $Q(\alpha' I)$,*

$$\int_{\mathbb{R}} \phi(t)\, (-w')(t)\, dt \;\le\; C_{\mathrm{test}}(\psi,\varepsilon,\alpha') \Big(\iint_{Q(\alpha' I)} |\nabla U|^2\, \sigma\, dt\, d\sigma \Big)^{1/2} \tag{6}$$

with $C_{\mathrm{test}}(\psi,\varepsilon,\alpha') := C_{\mathrm{rem}}(\alpha',\psi)\, \mathcal{A}_{\mathrm{adm}}(\psi,\varepsilon,\alpha')$ independent of I, t_0, L. In particular, defining the box-energy constant

$$C_{\mathrm{box}}^{(\zeta)} \;:=\; \sup_I\, \frac{1}{|I|} \iint_{Q(\alpha' I)} |\nabla U|^2\, \sigma\, dt\, d\sigma,$$

one has the scale bound

$$\int_{\mathbb{R}} \phi\, (-w') \;\le\; C_{\mathrm{test}}(\psi,\varepsilon,\alpha')\, \sqrt{C_{\mathrm{box}}^{(\zeta)}}\, L^{1/2}.$$

Proof. Apply Lemma 28 with $\phi \in \mathcal{W}_{\mathrm{adm}}(I;\varepsilon)$ and absorb the window-side constants into $C_{\mathrm{test}}(\psi,\varepsilon,\alpha')$. □

Lemma 30 (Whitney–uniform wedge)**.** *Fix parameters $\alpha' > 1$ and $\varepsilon \in (0, \frac{1}{4}]$. Fix the Whitney schedule and clip by $L_\star$: set $L_\star := c/\log 2$ and henceforth*

$$L(T) \;:=\; \min\Big\{ \frac{c}{\log\langle T\rangle},\; L_\star \Big\}.$$

Then for every Whitney interval $I = [t_0 - L, t_0 + L]$ and the corresponding cutoff $\psi_{L,t_0}(t) := \psi((t-t_0)/L) = Z_0 L\, \varphi_{L,t_0}(t)$ (so $\psi_{L,t_0} \equiv 1$ on I),

$$\int_{\mathbb{R}} \psi_{L,t_0}(t)\, (-w'(t))\, dt \;\le\; Z_0\, L_\star \cdot C_{\mathrm{test}}(\psi,\varepsilon,\alpha')\, \sqrt{C_{\mathrm{box}}^{(\zeta)}}\, L_\star^{1/2} \;:=\; \pi\, \Upsilon_{\mathrm{Whit}}(c).$$

Choosing $c > 0$ sufficiently small so that $\Upsilon_{\mathrm{Whit}}(c) < \frac{1}{2}$ yields the hypothesis of Lemma 30 and hence (P+)*.*

Proof. Since $\psi_{L,t_0} = Z_0 L\, \varphi_{L,t_0}$, apply Proposition 29 with $\phi = \varphi_{L,t_0}$, then multiply the resulting bound by $Z_0 L$ and use the Whitney clip $L \le L_\star$. □

Theorem 31 (Proof of Theorem 9)**.** *The boundary wedge* (P+) *holds for $\mathcal{J}_{\mathrm{out}}$.*

Proof. By the definition (4) and Theorem 21, the quantitative phase–velocity identity (Theorem 21) applies to the ζ-normalized unimodular ratio J_ζ, and hence (by (4)) to $\mathcal{J}_{\mathrm{out}}$. In particular, the associated boundary phase distribution $-w'$ is positive.

Proposition 29 (CR–Green pairing) supplies a uniform Whitney-scale bound for the windowed phase derivative in terms of the box energy $C_{\mathrm{box}}^{(\zeta)}$. Applying the Whitney schedule and choosing $c > 0$ small enough gives $\Upsilon_{\mathrm{Whit}}(c) < \frac{1}{2}$ in Lemma 30. Lemma 30 then yields (P+). □

B Supplementary computational verification protocol (verifier commands and expected fields)

There are two verification modes:

- **Verification using shipped artifacts:** verify the shipped JSON artifacts match Table 1.
- **Optional regeneration check (supplementary):** rerun the verifier to regenerate the artifacts from scratch.

Prerequisites

From the bundle's `compute/` directory, install the ARB/ball-arithmetic bindings:

```
pip install -r requirements.txt
```

Verification using shipped artifacts: check shipped JSON artifacts

- **Rectangle artifact** `artifacts/theta_certify_sigma06_07_t0_20_outer_zeta_proj.json`. Check (at minimum):
  - `results.ok = true`
  - `results.theta_hi = 0.9999928763...  < 1`
  - `results.processed_boxes = 380764`
- **Pick artifact** `artifacts/pick_sigma0599_raw_zeta_N16.json`. Check (at minimum):
  - `pick.delta_cert = 0.594...`
  - `pick.P_spd_at_0 = true`
  - `pick.tail_l1_partial_hi` (diagnostic L1 tail sum)
- **Pick artifact** `artifacts/pick_sigma06_raw_zeta_N16.json`. Check (at minimum):
  - `pick.delta_cert = 0.594...`
  - `pick.P_spd_at_0 = true`
  - `pick.tail_l1_partial_hi` (diagnostic L1 tail sum)
- **Pick artifact** `artifacts/pick_sigma07_raw_zeta_N16.json`. Check (at minimum):
  - `pick.delta_cert = 0.627...`
  - `pick.P_spd_at_0 = true`
  - `pick.tail_l1_partial_hi` (diagnostic L1 tail sum)

Optional regeneration check (supplementary): exact command lines

Run the verifier from the bundle's `compute/` directory (or the corresponding directory in a repository checkout). The following commands reproduce the primary artifacts (line breaks are for readability):

1) Rectangle certification (`theta_certify`).

```
python verify_attachment_arb.py \
  --theta-certify \
  --arith-gauge outer_zeta_proj \
  --arith-P-cut 2000 \
  --rect-sigma-min 0.6 --rect-sigma-max 0.7 \
  --rect-t-min 0.0 --rect-t-max 20.0 \
  --outer-mode midpoint \
  --outer-P-cut 2000 \
  --outer-T 50.0 --outer-n 2001 \
  --theta-init-n-sigma 10 --theta-init-n-t 50 \
  --theta-min-sigma-width 0.0001 --theta-min-t-width 0.001 \
  --theta-max-boxes 500000 \
  --prec 260 \
  --theta-out artifacts/theta_certify_sigma06_07_t0_20_outer_zeta_proj.json \
  --progress
```

2) Pick certification at $\sigma_0 = 0.599$ (`pick_certify`).

```
python verify_attachment_arb.py \
  --pick-certify \
  --pick-sigma0 0.599 \
  --pick-N 16 \
  --pick-coeff-count 128 \
  --pick-K 512 \
  --pick-rho 0.4 \
  --pick-rho-bound 0.5 \
  --arith-gauge raw_zeta \
  --arith-P-cut 2000 \
  --prec 1024 \
  --pick-out artifacts/pick_sigma0599_raw_zeta_N16.json
```

3) Pick certification at $\sigma_0 = 0.6$ (`pick_certify`).

```
python verify_attachment_arb.py \
  --pick-certify \
  --pick-sigma0 0.6 \
  --pick-N 16 \
  --pick-coeff-count 128 \
  --pick-K 512 \
  --pick-rho 0.4 \
  --pick-rho-bound 0.5 \
  --arith-gauge raw_zeta \
  --arith-P-cut 2000 \
  --prec 1024 \
  --pick-out artifacts/pick_sigma06_raw_zeta_N16.json
```

4) Pick certification at $\sigma_0 = 0.7$ (`pick_certify`).

```
python verify_attachment_arb.py \
  --pick-certify \
  --pick-sigma0 0.7 \
  --pick-N 16 \
  --pick-coeff-count 128 \
  --pick-K 512 \
  --pick-rho 0.4 \
  --pick-rho-bound 0.5 \
  --arith-gauge raw_zeta \
  --arith-P-cut 2000 \
  --outer-mode rigorous \
  --outer-P-cut 2000 \
  --prec 1024 \
  --pick-out artifacts/pick_sigma07_raw_zeta_N16.json
```

Meaning of successful verification

The verifier uses *ball arithmetic*: each computed quantity is an interval enclosure (midpoint plus radius) and every operation propagates rounding error outward. Thus each check is a formal inequality of the form `upper bound < 1` or `directed-rounding` $\mathrm{LDL}^\top$ `succeeds with positive pivots`. If the verification checks above pass, then the numerical inequalities summarized in Table 1 are certified within the logic of ball arithmetic.

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